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# QUIZ # 3
## (Theory of Computation)

A1: a) $\Sigma = \{0,1\}$

$L = \{w \in \Sigma^* \mid w \text{ is a palindrome}\}$

Hence L is of the form 000, 010, 1001, etc

$w = w^R$

The productions to generate such a language can be written as follows:

$$S \rightarrow 0S0 \mid 1S1 \mid 1 \mid 0 \mid \varepsilon$$

b) $L^c = \Sigma^* - L$

$\Rightarrow L^c = \{w \in \Sigma^* \mid w \text{ is NOT a palindrome}\}$

So, we need to write our productions in such a way that $w \neq w^R$

The productions to generate such a language can be written as follows:

$$\boxed{\begin{aligned} S &\rightarrow 0S0 \mid 1S1 \mid T \\ T &\rightarrow 1U0 \mid 0U1 \\ U &\rightarrow 1U \mid 2U \mid 0 \mid 1 \mid \varepsilon \end{aligned}}$$

Note that this set of productions forces us to have one production of the form 1U0 or 0U1 and this is sufficient to ensure that the string is non-palindromic.

So here we have obtained a CFG for $L^c$, hence we can say that $L^c$ is CFL.

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A2: $S \rightarrow XY0Z$

$X \rightarrow YZ$

$Y \rightarrow 1 \mid \varepsilon$

$Z \rightarrow W \mid \varepsilon$

$W \rightarrow 2$

Using step 2 of CNF:

(i) Remove $Y \rightarrow \varepsilon$

We add

$S \rightarrow X0Z$

$X \rightarrow Z$

to the productions

P: $S \rightarrow XY0Z \mid X0Z$

$X \rightarrow YZ \mid Z$

$Y \rightarrow 1$

$Z \rightarrow W \mid \varepsilon$

$W \rightarrow 2$

(ii) Remove $Z \rightarrow \varepsilon$

We add

$S \rightarrow XY0$

$S \rightarrow X0$

$X \rightarrow Y$

$X \rightarrow \varepsilon$

to the productions

P: $S \rightarrow XY0Z \mid X0Z \mid XY0 \mid X0$

$X \rightarrow YZ \mid Y \mid Z \mid \varepsilon$

$Y \rightarrow 1$

$Z \rightarrow W$

$W \rightarrow 2$

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(iii) Remove $X \to \varepsilon$

We add

$S \to XOZ$

$S \to OZ$

$S \to YO$

$S \to O$

to the productions:

$$\boxed{\begin{array}{ll} P: & S \to XYOZ \mid XOZ \mid XYO \mid XO \mid YOZ \mid OZ \mid YO \mid O \\ & X \to YZ \mid Y \mid Z \\ & Y \to 1 \\ & Z \to W \\ & W \to 2 \end{array}}$$

$\Rightarrow$ This grammar is equivalent to $G$, and has no $\varepsilon$-productions.

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A3: $\Sigma = \{0, 1, 2\}$

$L = \{w \in \Sigma^* \mid n_0(w) < n_1(w) \text{ and } n_0(w) < n_2(w)\}$

Let $w = 0^p 1^{2p} 2^{4p}$ (constraint is satisfied)

$p \to$ pumping const.

We know, for pumping lemma

$w = uvxyz$

i) $|vxy| \leq p$

ii) $|vy| > 0$

iii) $uv^ixy^iz \in L, \ \forall i \geq 0$

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Possibilities:

1) vxy contains only 0s
2) vxy contains only 1s
3) vxy contains only 2s
4) vxy straddles 1s and 2s
5) vxy straddles 0s and 1s

Note that vxy cannot contain all three symbols because $|vxy| \leq p$

★ In case 1) and 5), pumping up may violate the condition $n_0(w) < n_2(w)$

★ In case 2) and 3) and 4), pumping down may reduce the 1s or 2s s.t. at least one of the conditions $n_0(w) < n_1(w)$ or $n_0(w) < n_2(w)$ are violated.

Thus, from the pumping lemma, we can conclude that pumping in any scenario yields the string to not belong to L.
$\Rightarrow$ L is <u>not</u> context-free.

b) One feature that cannot be captured by a CFG →
Checking whether a variable has been declared before its use.
This can be considered to be similar to the language $L = wcw$ where the first w denotes the declaration and the second one is its first usage.
Note that just satisfying this condition doesn't guarantee that the variable has been declared. But if it has been declared, then this holds.
It can be shown that $L = \{wcw\}$ is not CF and hence, this feature cannot be captured by a CFG.

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